

3 Demonstrates how to simulate a Data Center with one host and run one Cloudlet on it using cloudsimsim.

Code Implementation:

```
package org.cloudbus.cloudsim.examples;

import java.text.DecimalFormat;
import java.util.*;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class CloudSimExample1 {
    static List<Cloudlet> cl=new ArrayList<>();
    static List<Vm> vm=new ArrayList<>();

    public static void main(String[] a){
        Log.println("Starting CloudSimExample1...");
        try{
            CloudSim.init(1,Calendar.getInstance(),false);

            Datacenter d=createDC("Datacenter_0");
            DatacenterBroker b=new DatacenterBroker("Broker");
            int id=b.getId();

            vm.add(new Vm(0,id,1000,1,512,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            b.submitVmList(vm);

            UtilizationModel u=new UtilizationModelFull();
            Cloudlet c=new Cloudlet(0,400000,1,300,300,u,u,u);
            c.setUserId(id); c.setVmId(0);

            cl.add(c);
            b.submitCloudletList(cl);

            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            print(b.getCloudletReceivedList());
            d.printDebts();

            Log.println("CloudSimExample1 finished!");
        } catch (Exception e) {e.printStackTrace();}
    }

    static Datacenter createDC(String n){
        List<Pe> p=new ArrayList<>();
        p.add(new Pe(0,new PeProvisionerSimple(1000)));

        List<Host> h=new ArrayList<>();
        h.add(new Host(0,new RamProvisionerSimple(2048),
```

```

new BwProvisionerSimple(10000),1000000,p,new VmSchedulerTimeShared(p));

DatacenterCharacteristics c=new DatacenterCharacteristics(
"x86","Linux","Xen",h,10.0,3.0,0.05,0.001,0.0);

try {
return new Datacenter(n,c,new VmAllocationPolicySimple(h),
new LinkedList<Storage>(),0);
} catch(Exception e){return null;}
}

static void print(List<Cloudlet> l){
Log.println("\n===== OUTPUT =====");
Log.println("Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time   Finish
Time");
DecimalFormat f=new DecimalFormat("###.###");

for(Cloudlet c:l)
if(c.getCloudletStatus()==Cloudlet.SUCCESS){
Log.print("   "+c.getCloudletId()+"   SUCCESS");
Log.println("       "+c.getResourceId()
+"       "+c.getVmId()
+"       "+f.format(c.getActualCPUTime())
+"       "+f.format(c.getExecStartTime())
+"       "+f.format(c.getFinishTime()));
}
}
}

```

- 4 Demonstrates how to create a datacenter with one host, two virtual machines and run two cloudlets on it. Both virtual machine (vm1,vm2) has same machine configuration.

Code Implementation:

```
package org.cloudbus.cloudsim.examples;

import java.text.DecimalFormat;
import java.util.*;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class CloudSimExample2 {
    static List<Cloudlet> cl=new ArrayList<>();
    static List<Vm> vm=new ArrayList<>();

    public static void main(String[] a){
        Log.println("Starting CloudSimExample2...");
        try{
            CloudSim.init(1,Calendar.getInstance(),false);

            Datacenter d=createDC("Datacenter_0");
            DatacenterBroker b=new DatacenterBroker("Broker");
            int id=b.getId();

            vm.add(new Vm(0,id,250,1,512,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            vm.add(new Vm(1,id,250,1,512,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            b.submitVmList(vm);

            UtilizationModel u=new UtilizationModelFull();
            Cloudlet c1=new Cloudlet(0,250000,1,300,300,u,u,u);
            c1.setUserId(id);
            Cloudlet c2=new Cloudlet(1,250000,1,300,300,u,u,u);
            c2.setUserId(id);
            cl.add(c1); cl.add(c2);
            b.submitCloudletList(cl);

            b.bindCloudletToVm(0,0);
            b.bindCloudletToVm(1,1);

            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            print(b.getCloudletReceivedList());
            d.printDebts();

            Log.println("CloudSimExample2 finished!");
        } catch (Exception e) {e.printStackTrace();}
    }

    static Datacenter createDC(String n){
```

```

List<Pe> p=new ArrayList<>();
p.add(new Pe(0,new PeProvisionerSimple(1000)));
List<Host> h=new ArrayList<>();
h.add(new Host(0,new RamProvisionerSimple(2048),
new BwProvisionerSimple(10000),1000000,p,new VmSchedulerTimeShared(p)));

DatacenterCharacteristics c=new DatacenterCharacteristics(
"x86","Linux","Xen",h,10.0,3.0,0.05,0.001,0.0);

try{return new Datacenter(n,c,new VmAllocationPolicySimple(h),new LinkedList<Storage>(),0);}
catch(Exception e){return null;}
}

static void print(List<Cloudlet> l){
Log.println("\n===== OUTPUT =====");
Log.println("Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time   Finish
Time");
DecimalFormat f=new DecimalFormat("###.##");

for(Cloudlet c:l)
if(c.getCloudletStatus()==Cloudlet.SUCCESS){
Log.print("   "+c.getCloudletId()+"   SUCCESS");
Log.println("       "+c.getResourceId()+"       "+c.getVmId()+
"       "+f.format(c.getActualCPUTime())+
"       "+f.format(c.getExecStartTime())+
"       "+f.format(c.getFinishTime()));
}
}
}
}

```

5

Demonstrates how to create a datacenter with two hosts, two virtual machines and run two cloudlets on it. The cloudlets run in VMs with different MIPS requirements. The second VM will have twice the priority of virtual machine one (VM1) and so cloudlet will receive twice CPU time to complete the execution.

Code Implementation:

```
package org.cloudbus.cloudsim.examples;

import java.text.DecimalFormat;
import java.util.*;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class CloudSimExample3 {
    static List<Cloudlet> cl=new ArrayList<>();
    static List<Vm> vm=new ArrayList<>();

    public static void main(String[] a){
        Log.println("Starting CloudSimExample3...");
        try{
            CloudSim.init(1,Calendar.getInstance(),false);

            Datacenter d=createDC("Datacenter_0");
            DatacenterBroker b=new DatacenterBroker("Broker");
            int id=b.getId();

            vm.add(new Vm(0,id,250,1,2048,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            vm.add(new Vm(1,id,500,1,2048,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            b.submitVmList(vm);

            UtilizationModel u=new UtilizationModelFull();
            Cloudlet c1=new Cloudlet(0,40000,1,300,300,u,u,u);
            c1.setUserId(id);
            Cloudlet c2=new Cloudlet(1,40000,1,300,300,u,u,u);
            c2.setUserId(id);
            cl.add(c1); cl.add(c2);
            b.submitCloudletList(cl);

            b.bindCloudletToVm(0,0);
            b.bindCloudletToVm(1,1);

            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            print(b.getCloudletReceivedList());
            d.printDebts();

            Log.println("CloudSimExample3 finished!");
        } catch (Exception e) {e.printStackTrace();}
```

```

}

static Datacenter createDC(String n){
List<Pe> p1=new ArrayList<>(); p1.add(new Pe(0,new PeProvisionerSimple(1000)));
List<Pe> p2=new ArrayList<>(); p2.add(new Pe(0,new PeProvisionerSimple(1000)));

List<Host> h=new ArrayList<>();
h.add(new Host(0,new RamProvisionerSimple(2048),new
BwProvisionerSimple(10000),1000000,p1,new VmSchedulerTimeShared(p1)));
h.add(new Host(1,new RamProvisionerSimple(2048),new
BwProvisionerSimple(10000),1000000,p2,new VmSchedulerTimeShared(p2)));

DatacenterCharacteristics c=new DatacenterCharacteristics(
"x86","Linux","Xen",h,10.0,3.0,0.05,0.001,0.0);

try{return new Datacenter(n,c,new VmAllocationPolicySimple(h),new LinkedList<Storage>(),0);}
catch(Exception e){return null;}
}

static void print(List<Cloudlet> l){
Log.println("\n===== OUTPUT =====");
Log.println("Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time   Finish
Time");
DecimalFormat f=new DecimalFormat("###.###");
for(Cloudlet c:l)
if(c.getCloudletStatus()==Cloudlet.SUCCESS){
Log.print("   "+c.getCloudletId()+"   SUCCESS");
Log.println("       "+c.getResourceId()+"       "+c.getVmId()+
"       "+f.format(c.getActualCPUTime())+
"       "+f.format(c.getExecStartTime())+
"       "+f.format(c.getFinishTime()));
}
}
}
}

```

- 6 Demonstrate how to create two datacenters with one host each and run two cloudlets on them.
Code Implementation:

```
package org.cloudbus.cloudsim.examples;

import java.text.DecimalFormat;
import java.util.*;
import org.cloudbus.cloudsim.*;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.*;

public class CloudSimExample4 {
    static List<Cloudlet> cl=new ArrayList<>();
    static List<Vm> vm=new ArrayList<>();

    public static void main(String[] a){
        Log.println("Starting CloudSimExample4...");
        try{
            CloudSim.init(1,Calendar.getInstance(),false);

            Datacenter d0=createDC("Datacenter_0");
            Datacenter d1=createDC("Datacenter_1");
            DatacenterBroker b=new DatacenterBroker("Broker");
            int id=b.getId();

            vm.add(new Vm(0,id,250,1,512,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            vm.add(new Vm(1,id,250,1,512,1000,10000,"Xen",new CloudletSchedulerTimeShared()));
            b.submitVmList(vm);

            UtilizationModel u=new UtilizationModelFull();
            Cloudlet c1=new Cloudlet(0,40000,1,300,300,u,u,u); c1.setUserId(id);
            Cloudlet c2=new Cloudlet(1,40000,1,300,300,u,u,u); c2.setUserId(id);
            cl.add(c1); cl.add(c2);
            b.submitCloudletList(cl);

            b.bindCloudletToVm(0,0);
            b.bindCloudletToVm(1,1);

            CloudSim.startSimulation();
            CloudSim.stopSimulation();

            print(b.getCloudletReceivedList());
            d0.printDebts();
            d1.printDebts();

            Log.println("CloudSimExample4 finished!");
        } catch(Exception e){e.printStackTrace();}
    }
}
```

```

static Datacenter createDC(String n){
List<Pe> p=new ArrayList<>(); p.add(new Pe(0,new PeProvisionerSimple(1000)));
List<Host> h=new ArrayList<>();
h.add(new Host(0,new RamProvisionerSimple(2048),new
BwProvisionerSimple(10000),1000000,p,new VmSchedulerSpaceShared(p)));
DatacenterCharacteristics c=new DatacenterCharacteristics(
"x86","Linux","Xen",h,10.0,3.0,0.05,0.001,0.0);
try{return new Datacenter(n,c,new VmAllocationPolicySimple(h),new LinkedList<Storage>(),0);}
catch(Exception e){return null;}
}

static void print(List<Cloudlet> l){
Log.println("\n===== OUTPUT =====");
Log.println("Cloudlet ID   STATUS   Data center ID   VM ID   Time   Start Time   Finish
Time");
DecimalFormat f=new DecimalFormat("###.##");
for(Cloudlet c:l)
if(c.getCloudletStatus()==Cloudlet.SUCCESS){
Log.print("   "+c.getCloudletId()+"   SUCCESS");
Log.println("       "+c.getResourceId()+"       "+c.getVmId()+
"       "+f.format(c.getActualCPUTime())+
"       "+f.format(c.getExecStartTime())+
"       "+f.format(c.getFinishTime()));
}
}
}
}

```